



Fuel from humans

Researchers at PNNL developed a new method to produce biofuel from human excrements. <http://dgit.in/BiofuelHE>



Verizon fiber connection

Verizon begins rolling out its new \$300 million FiOS fiber connection in Boston, US. Learn more about it here: <http://dgit.in/VrznFbr>

nology. The advent of photo manipulation tools raised similar concerns but that hasn't screwed up the legal system and we don't expect this to either.

Adobe has themselves said that they're working to detect use of its software so if at all any fraudulent evidence is presented, they'll know if it was made using VoCo technology.

They can learn on their own

We've seen how far machine learning algorithms have come and how the hardware advances, especially the GPUs, have really pushed the boundaries of machine learning to new extremes. However, one thing remains quite constant, a human expert is always required to perform certain key tasks that set a machine learning system on its path. The human element is required to preprocess the data, select the key parameters, select the most apt ML algorithm for the learning to begin, tweak the system to ensure accuracy during the training phase and lastly, to precisely analyse if the system is doing what it set out to do.

With AutoML, you can say goodbye to the humans. This rapid growth of ML in the last few years has led to an increased demand for ML solutions that work out of the box requiring next to no expertise to set up. AutoML does exactly that. It provides an array of tools that any novice can apply on a set of data – from simple metadata learning to traditional algorithms that use Bayesian classification. It won't be long before complex AI can finally build and improve itself over time.

They can keep secrets

Another one from the folks at Google Brain, this experiment involved setting up three neural networks which could all communicate with each other. Two of them, Bob and Alice, shared a secret key which was used to encrypt communication between the two. And the third neural network, called Eve, was the snoop who'd try to eavesdrop on the conversation between Bob and Alice. Eve's success was measured by how accurate her interpretation of the

secret communication was. And Alice and Bob's success was measured with reference to how less of their conversation could be decrypted by Eve.

This resulted in an adversarial network between the three entities with improvements happening over time. Bob and Alice became very good at generating an inhumanly powerful encryption that Eve could not crack. That's right, robots just created encryption methods that even other powerful robots could not crack. The techniques by which these new methods were arrived at are way advanced than how humans have done so far.

All aboard the Open Source train

2016 was a huge year for AI and quite a lot of these have gone the open source way. Google announced the open sourcing of Tensor Flow and started supporting Keras, a deep learning library for Python. Microsoft then went ahead and did the same with



Google opensourced TensorFlow

CNTK (Cognitive Tool Kit), their set of learning algorithms. The Chinese web-services company, Baidu, also open sourced their libraries under the name PaddlePaddle. Amazon has opted to implement MXNet in their AWS ML platform and lastly, Facebook is backing Torch and Caffe.

2017 will be bigger

While 2016 has already accomplished so much, 2017 looks to be even bigger with leaps of advancement being predicted in every single domain. But two things stand out – chatbots and self-driving automobiles. While there

have been a few setbacks for chatbots this year, namely, Microsoft's Tay going on racist rants thanks for hundreds of thousands of Twitter users feeding it racist diatribe, the year 2017 seems to be a lot more promising. Google's Allo, Amazon's Alexa, Apple's Siri are all paving the way forward with better interpretation of natural requests and responses becoming more intuitive and natural. Given the rapid pace of improvements, we hope to see a lot more human-like chatbots in 2017.

Self-driving automobiles are doing well enough but there have been a few legal setbacks and they're still to gain mainstream acceptance. 2017 should probably be the year when public transportations systems gain a level of autonomy post. The consumer segment should follow right behind but given the push back from the traditional automotive industry, this could take a few more years. Nevertheless, 2017 will see a huge leap in at least industrial applications for the self-driving vehicle.

Closer to the singularity

One of the prime reasons why Stephen Hawking and Elon Musk are worried is that once AI achieves the ability to communicate secretly and teach itself – It will give rise to a runaway effect resulting in a super intelligence. Mr. Hawking, in an interview with the BBC said that, "It would take off on its own, and redesign itself at an ever increasing rate. Humans, who are limited by slow biological evolution, couldn't compete, and would be superseded."

Elon Musk on the other hand is worried about just one company which he hasn't named yet, but he decided to do something about it. And thus, OpenAI was born. It's a non-profit AI company with an aim to develop friendly AI that hopefully doesn't end up like HAL 9000 from Arthur C. Clarke's Space Odyssey.

Whether we get superseded by machines as the dominant race or if we form a peaceful society where AI and humans can co-exist remains to be seen, but one thing's for certain, this will happen in our lifetime. 